



$\infty - \infty$?

$$\lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{1}{\sin x} \right) = \lim_{x \rightarrow 0} \left(\frac{\sin x - x}{x \sin x} \right) =$$
$$= \lim_{x \rightarrow 0} \frac{\cos x - 1}{\sin x + x \cos x} =$$

$$\lim_{x \rightarrow 0} \frac{-\sin x}{\cos x - \sin x + \cos x} =$$

$$= \lim_{x \rightarrow 0} \frac{-\sin x}{2\cos x - \sin x} = \lim_{x \rightarrow 0} \frac{0}{2} = 0$$

$$\lim_{x \rightarrow 0} \left(\frac{1}{x^2} - \frac{1}{\sin^2 x} \right)$$

$$\lim_{x \rightarrow 0} \frac{\sin^2 x - x^2}{x^2 \sin^2 x} = \lim_{x \rightarrow 0} \frac{2 \sin x \cdot \cos x - 2x}{2x \sin^2 x + 2 \sin x \cos x \cdot x^2}$$

$$\lim_{x \rightarrow 0} \frac{\sin 2x - 2x}{2x \sin^2 x + \sin 2x \cdot x^2} =$$

$$\lim_{x \rightarrow 0} \frac{\cos 2x \cdot 2 - 2}{2 \sin^2 x + 2x \cdot 2 \sin x \cos x + 2x \sin 2x + \cos 2x \cdot 2 \cdot x^2}$$

$$= \lim_{x \rightarrow 0}$$

$$\frac{-2 \sin 2x \cdot 2 \cdot 2}{2 \cdot 2 \sin x \cdot \cos x + 2x \cos 2x \cdot 2 + 2 \sin 2x}$$

$$+ 2 \sin 2x + 2 \sin 2x \cdot 2x +$$

$$+ 2 \cdot 2x \cdot \cos 2x + 2 \sin 2x \cdot x^2$$

$$= +\infty$$

$$-16 \cos 2x$$

$$= +\infty$$

$$= +\infty$$

$$\lim_{x \rightarrow 0} x^{\sin x} = \lim_{x \rightarrow 0} \ln(x^{\sin x}) =$$

$$\lim_{x \rightarrow 0} \sin x \ln x =$$

$$\lim_{x \rightarrow 0} \ln(x^{\sin x}) =$$

$$0 \cdot (-\infty)$$

$$0 \cdot (-\infty)$$

$$\lim_{x \rightarrow 0} \sin x \ln x =$$

$$\lim_{x \rightarrow 0} \sin x \ln x =$$

$$\lim_{x \rightarrow 0} \sin x = 0$$

$$\lim_{x \rightarrow 0} \ln x = -\infty$$

$$\lim_{x \rightarrow 0} \frac{\ln x}{\frac{1}{\sin x}} =$$

$$\frac{1}{\sin x} = \frac{1}{0} = +\infty$$

$$\lim_{x \rightarrow 0} \frac{\frac{1}{x}}{-\frac{\cos x}{\sin^2 x}} \quad \Leftrightarrow \quad \left(\frac{1}{\sin x} \right)' = \frac{-\cos x}{\sin^2 x}$$

$$\Leftrightarrow \lim_{x \rightarrow 0} \frac{1}{x} : \left(\frac{-\cos x}{\sin^2 x} \right) =$$

$$= \lim_{x \rightarrow 0} -\frac{1}{x} \cdot \frac{\sin^2 x}{\cos x} =$$

$$= \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right) \left(-\frac{\sin x}{\cos x} \right) =$$

$$= 1 \cdot \lim_{x \rightarrow 0} \left(-\frac{\sin x}{\cos x} \right)$$

$$\lim_{x \rightarrow 0} (1+2x)^{\frac{3}{x}} =$$

$$\lim_{x \rightarrow 0} \frac{3}{x} \ln(1+2x) =$$

$$= \lim_{x \rightarrow 0} \frac{3 \cdot \frac{1}{1+2x} \cdot 2}{1} = 6$$

$$\lim_{x \rightarrow \infty} \left(\frac{x+2}{x-1} \right)^x \quad \text{---} \quad \infty^\infty$$

$$\stackrel{L}{=} \lim_{x \rightarrow \infty} x \ln \left(\frac{x+2}{x-1} \right)$$

$$\stackrel{L}{=} \ln \left(\frac{x+2}{x-1} \right) = \ln \left(\frac{x-1+3}{x-1} \right)$$

$$= \ln \left(1 + \frac{3}{x-1} \right)$$

$$\infty \cdot 0$$

$$\stackrel{L}{=} \lim_{x \rightarrow \infty} x \cdot \ln \left(1 + \frac{3}{x-1} \right)$$

$$\stackrel{L}{=} \lim_{x \rightarrow \infty} \frac{\ln \left(1 + \frac{3}{x-1} \right)}{\frac{1}{x}} \quad \text{---} \quad \frac{0}{0}$$

$$\stackrel{L}{=} \frac{\frac{1}{1 + \frac{3}{x-1}} \cdot \left(-\frac{3}{(x-1)^2} \right)}{-\frac{1}{x^2}}$$

$$\stackrel{L}{=} \lim_{x \rightarrow \infty} \frac{1}{x^2}$$

$$\left(\frac{\cancel{(x-1)}}{\cancel{(x-1)} + 3} \right) \left(\frac{-3}{(x-1)^2} \right) =$$

$$\frac{-3}{(x-1)(x-1) + 3(x-1)}$$

$$= \frac{1}{x^2}$$

$$\lim_{x \rightarrow \infty} \frac{3x^2}{(x-1)^2 + 3(x-1)}$$

$$\lim_{x \rightarrow \infty} \frac{6x}{2(x-1) + 3} =$$

$$= \lim_{x \rightarrow \infty} \frac{6}{2} = \textcircled{3}$$

$$\lim_{x \rightarrow 0} \frac{\ln(1+x^2)}{1-\cos x} = \lim_{x \rightarrow 0} \frac{\frac{2x}{1+x^2}}{\sin x}$$

$$\lim_{x \rightarrow 0} \frac{2x}{(1+x^2)\sin x} =$$

$$= \lim_{x \rightarrow 0} \frac{2}{2x \sin x + (1+x^2) \cos x}$$

$$= \lim_{x \rightarrow 0} \frac{2}{0+1} = 2$$

$$\lim_{x \rightarrow 0} x \left(e^{\frac{1}{x}} - 1 \right) =$$

$$= \lim_{x \rightarrow 0} \frac{e^{\frac{1}{x}} - 1}{\frac{1}{x}} = \lim_{x \rightarrow 0} \frac{\cancel{\frac{1}{x^2}} \cdot e^{\frac{1}{x}}}{-\cancel{\frac{1}{x^2}}} =$$

$$\lim_{x \rightarrow 0} e^{\frac{1}{x}} = \infty$$

$$\lim_{x \rightarrow 0} x \ln x \quad (\Rightarrow) \quad 0 \cdot \infty$$

$$\begin{aligned} (\Rightarrow) \lim_{x \rightarrow 0} \frac{\ln x}{\frac{1}{x}} &= \lim_{x \rightarrow 0} \frac{\frac{1}{x}}{-\frac{1}{x^2}} = \\ &= \lim_{x \rightarrow 0} \frac{1}{x} (-x^2) = \lim_{x \rightarrow 0} -x = 0 \end{aligned}$$

$$\lim_{x \rightarrow 0} \frac{e^x - e^{-x} - 2x}{x - \sin x} =$$

$$\lim_{x \rightarrow 0} \frac{e^x + e^{-x} - 2}{1 - \cos x} = \frac{1+1-2}{1-1} = \frac{0}{0}$$

$$\lim_{x \rightarrow 0} \frac{e^x - e^{-x}}{\sin x} = \lim_{x \rightarrow 0} \frac{e^x + e^{-x}}{\cos x} = 2$$

$$\lim_{x \rightarrow \infty} \left(\frac{x}{x+1} \right)^x = \lim_{x \rightarrow \infty} \ln \left(\frac{x}{x+1} \right)^x$$

$$= e^{\lim_{x \rightarrow \infty} x \ln \left(\frac{x}{x+1} \right)}$$

$$\frac{x}{x+1} = \frac{x+1-1}{x+1} =$$

$$e^{\lim_{x \rightarrow \infty} \ln x \ln \left(1 - \frac{1}{x+1} \right)} =$$

$$\lim_{x \rightarrow \infty} \frac{\ln \left(1 - \frac{1}{x+1} \right)}{\frac{1}{x}} = \frac{0}{0}$$

$$\begin{aligned}
 & \lim_{x \rightarrow \infty} \frac{\frac{1}{1 - \frac{1}{x+1}} \cdot \left(\frac{1}{(x+1)^2} \right)}{-\frac{1}{x^2}} = \\
 & = \lim_{x \rightarrow \infty} \frac{\frac{\cancel{x+1}}{(x+1)-1} \cdot \frac{1}{(x+1)^2}}{-\frac{1}{x^2}} =
 \end{aligned}$$

$$\begin{aligned}
 & = \lim_{x \rightarrow \infty} \frac{\frac{1}{x(x+1)}}{-\frac{1}{x^2}} = \lim_{x \rightarrow \infty} \frac{-x^2}{x(x+1)} \\
 & = \lim_{x \rightarrow \infty} \frac{-x}{x+1} = \\
 & = \lim_{x \rightarrow \infty} \frac{-1}{1} = e^{-1}
 \end{aligned}$$

$$\begin{aligned}
 & \lim_{x \rightarrow 0} (\cos x)^{\frac{1}{x^2}} \quad \textcircled{=} \\
 & \textcircled{=} \lim_{x \rightarrow 0} \ln \cos x^{\frac{1}{x^2}} = \lim_{x \rightarrow 0} \frac{1}{x^2} \ln \cos x =
 \end{aligned}$$

$$= \lim_{x \rightarrow 0} \frac{\frac{1}{\cos x} \cdot (-\sin x)}{2x} = \frac{0}{0}$$

$$= \lim_{x \rightarrow 0} \frac{-\sin x}{2x \cos x} = \lim_{x \rightarrow 0} \frac{-\cos x}{2 \cos x - 2x \sin x}$$

$$= \lim_{x \rightarrow 0} \frac{-1}{2} = -\frac{1}{2}$$

$$\frac{1}{\sqrt{e}}$$